

Java SE 21 Developer (Exam 1Z0-830) Concurrent Programming

Introducing Java Concurrency

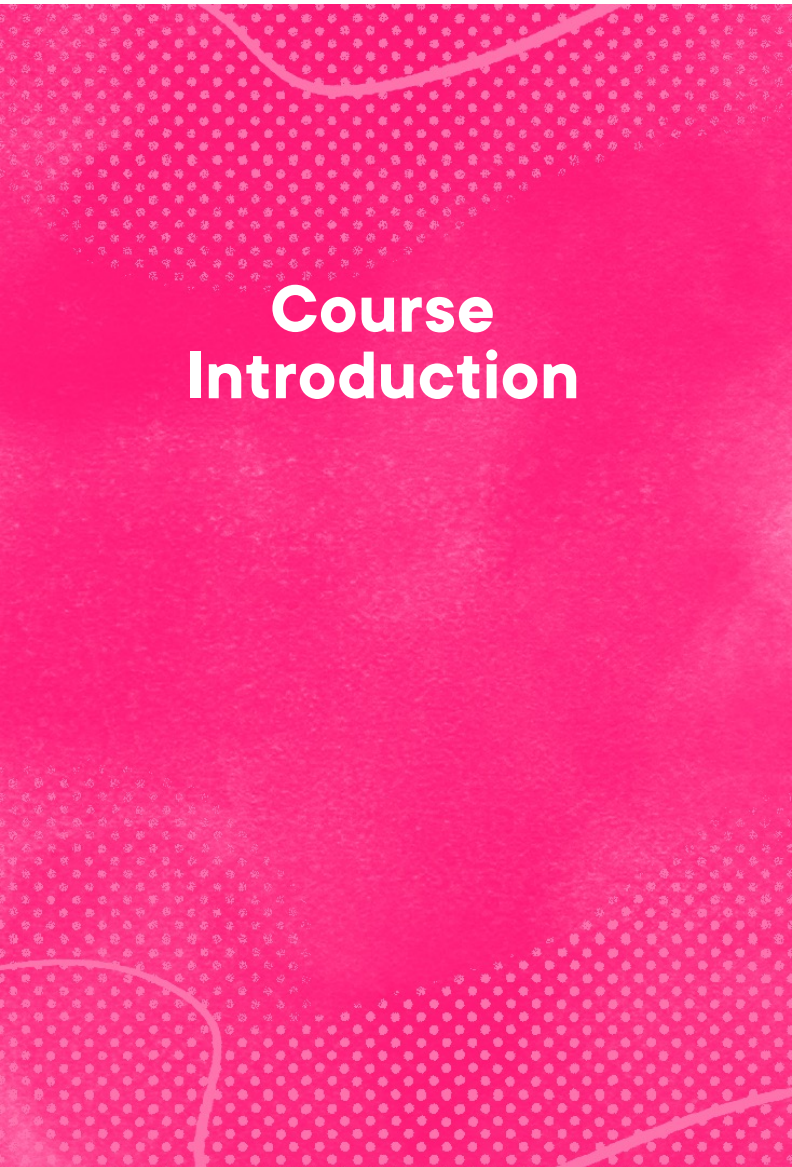


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Course Introduction

Examine the concurrency problem

Threads and their life cycle

The Runnable and Callable interfaces

Platform and virtual threads

Java Executors

Future objects

Synchronization mechanisms

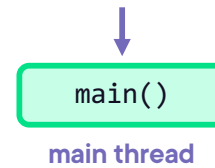
Concurrent collections

Parallel streams

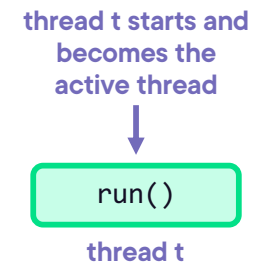
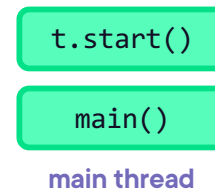


The Concurrency Problem

```
public static void main(String[] args) {  
    ...  
}
```



```
Runnable r = new Job();  
Thread t = new Thread(r);  
t.start();  
myMethod();
```



The JVM switches between the main thread and thread `t` until both threads complete.

the active thread again



Launching a New Thread

```
Runnable job = new MyRunnable();
```

```
Thread myThread = new Thread(job);
```

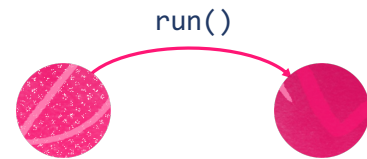
```
myThread.start();
```



Runnable object



Thread object

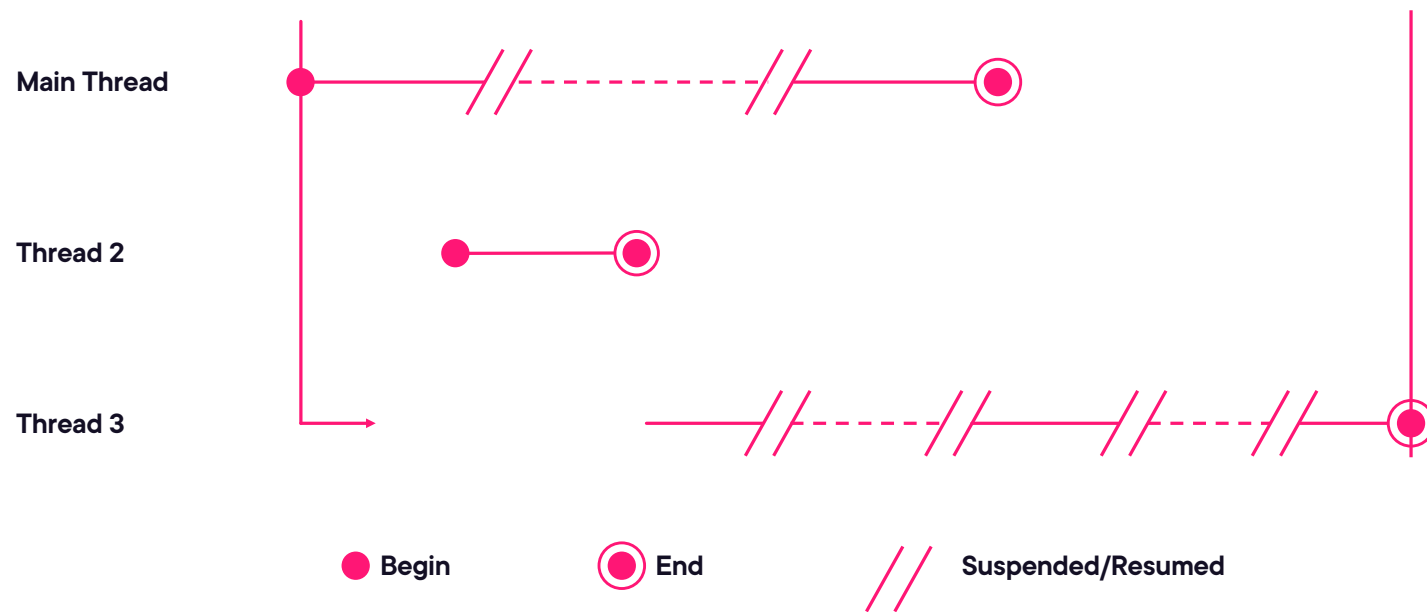


Thread object

Runnable object



Multithreading Mechanism



The Runnable Interface

```
@FunctionalInterface
public interface Runnable {
    /**
     * Runs this operation.
     */
    void run();
}
```



The Callable Interface

```
@FunctionalInterface
public interface Callable<V> {
    /**
     * Computes a result, or throws an
     * exception if unable to do so.
     *
     * @return computed result
     * @throws Exception if unable to
     * compute a result
     */
    V call() throws Exception;
}
```

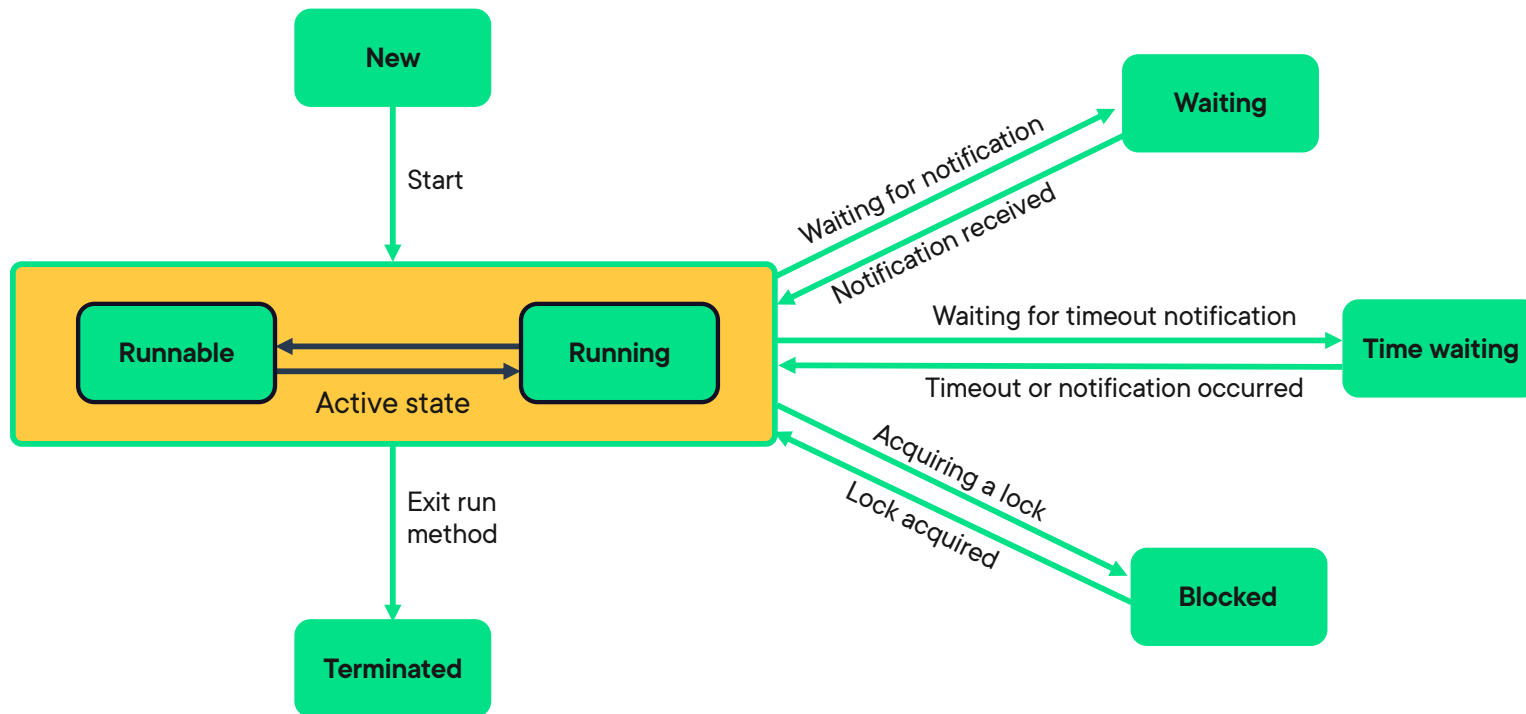


The Thread Class

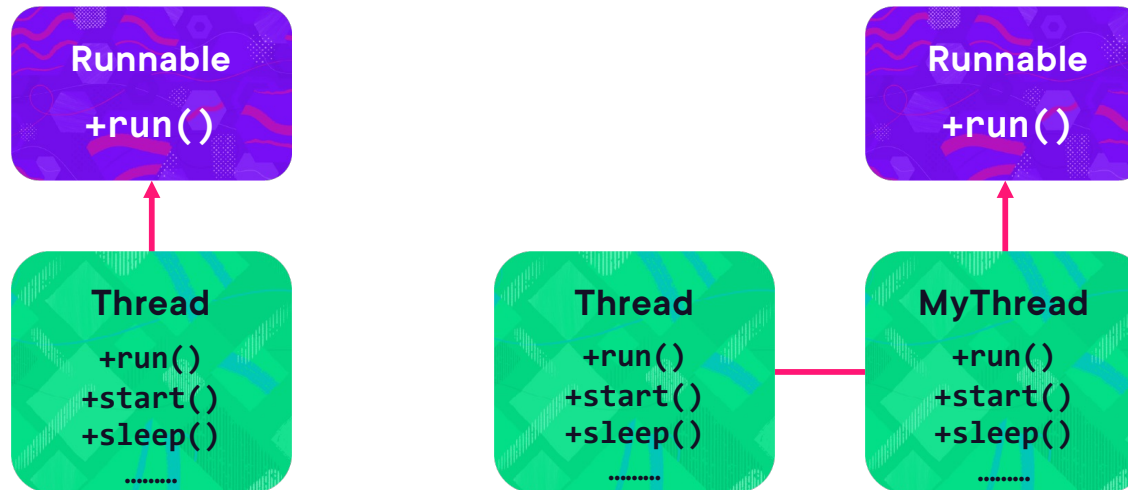
```
public class Thread implements Runnable {  
  
    public Thread();  
    public Thread(String name);  
    public Thread(Runnable target);  
    public Thread(Runnable target,  
                  String name);  
  
}
```



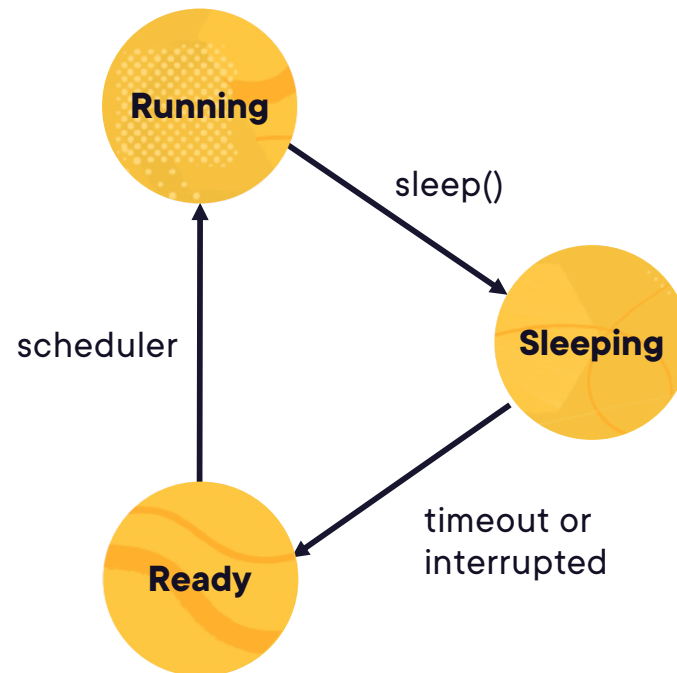
Life Cycle of Threads



Thread and Runnable



Thread.sleep() method





Demo: One Thread Demo





Demo: Two Threads Demo

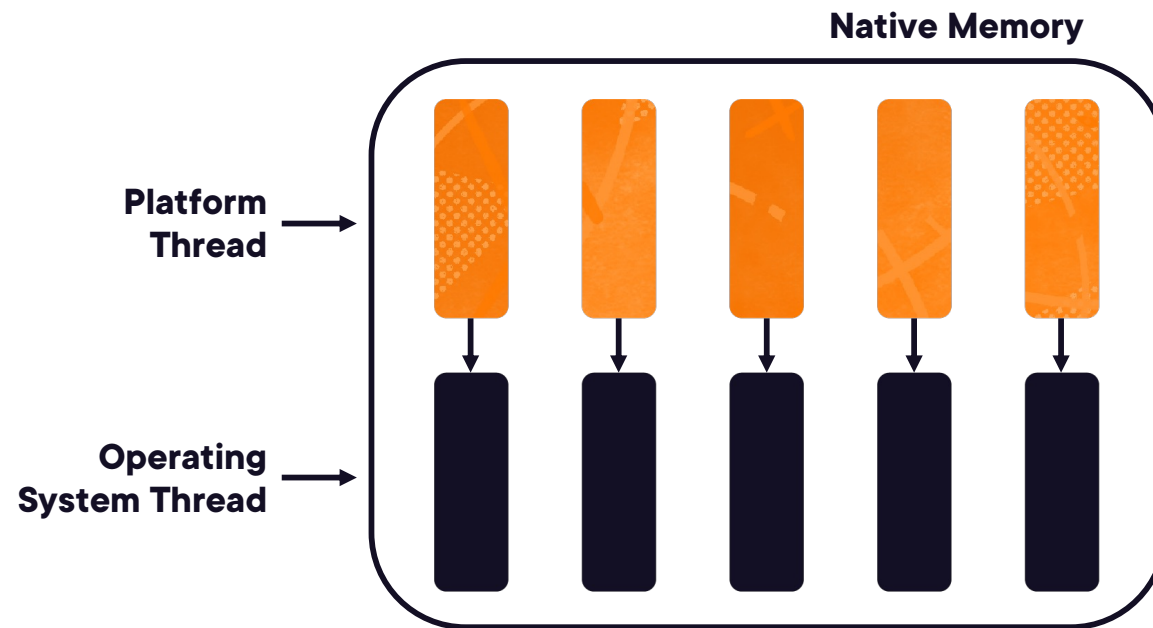




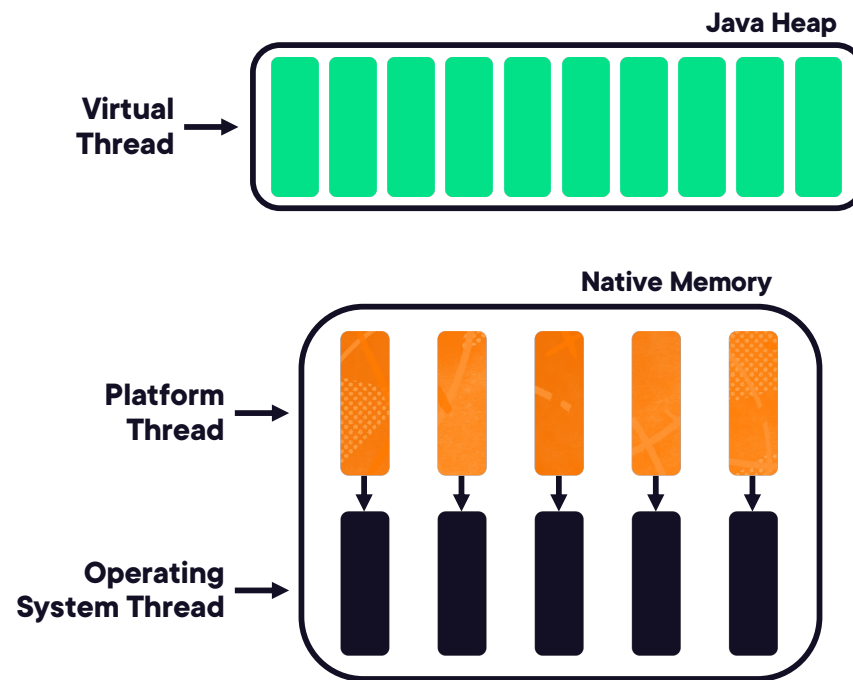
Virtual Threads



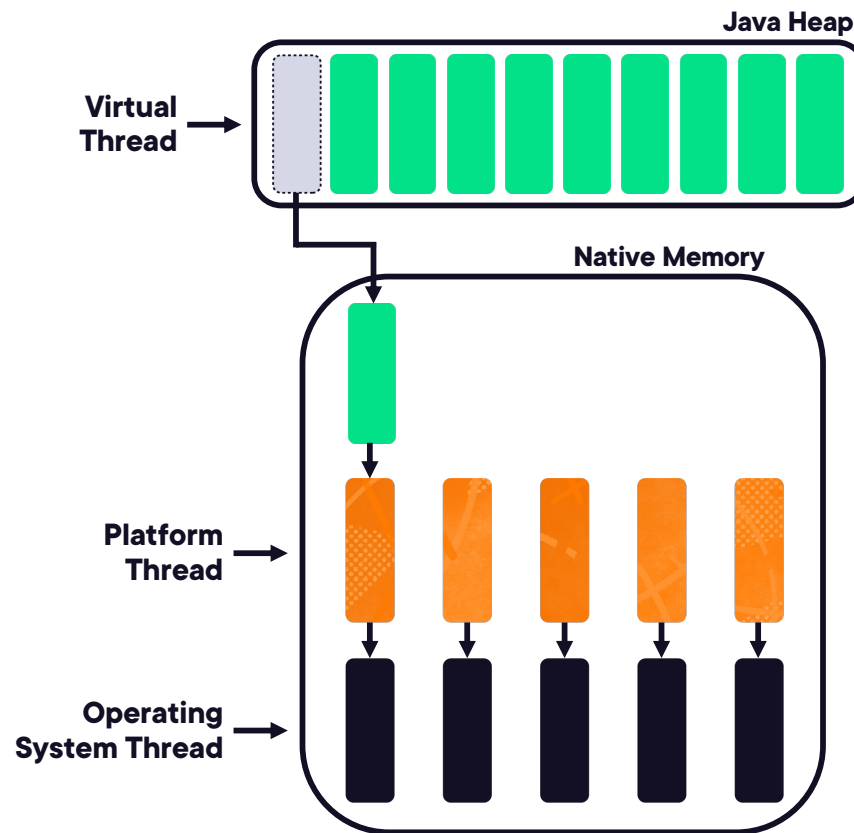
Platform Threads

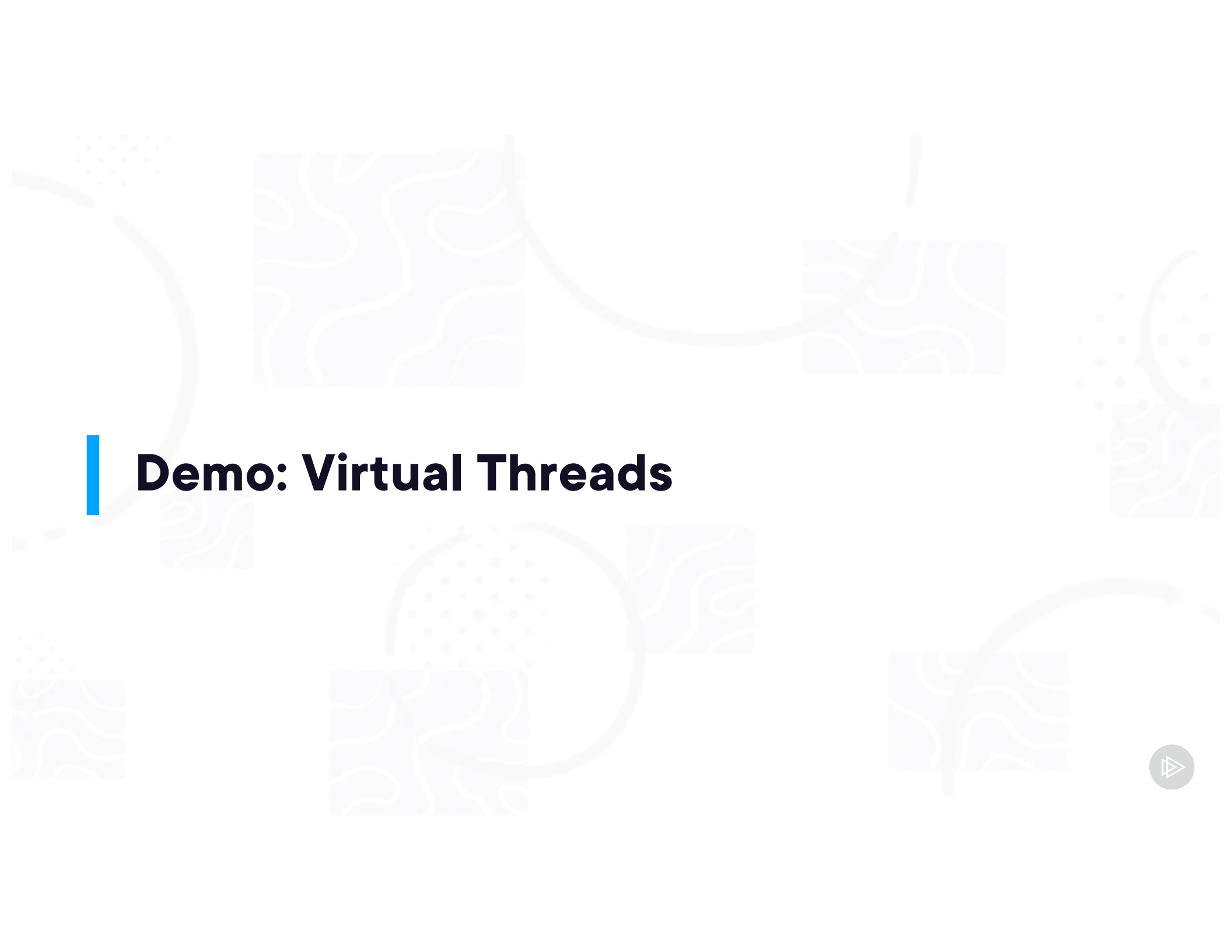


Virtual Threads



Working with Virtual Threads





Demo: Virtual Threads

